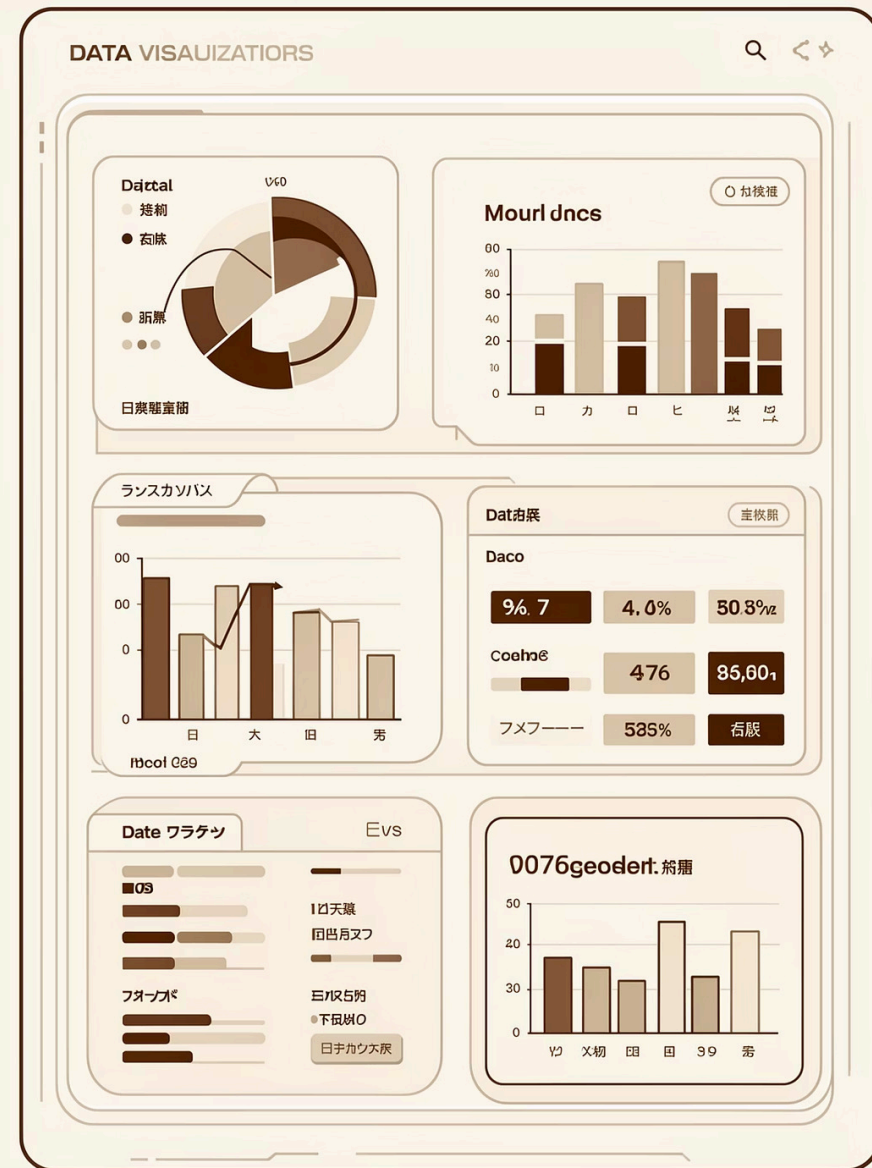
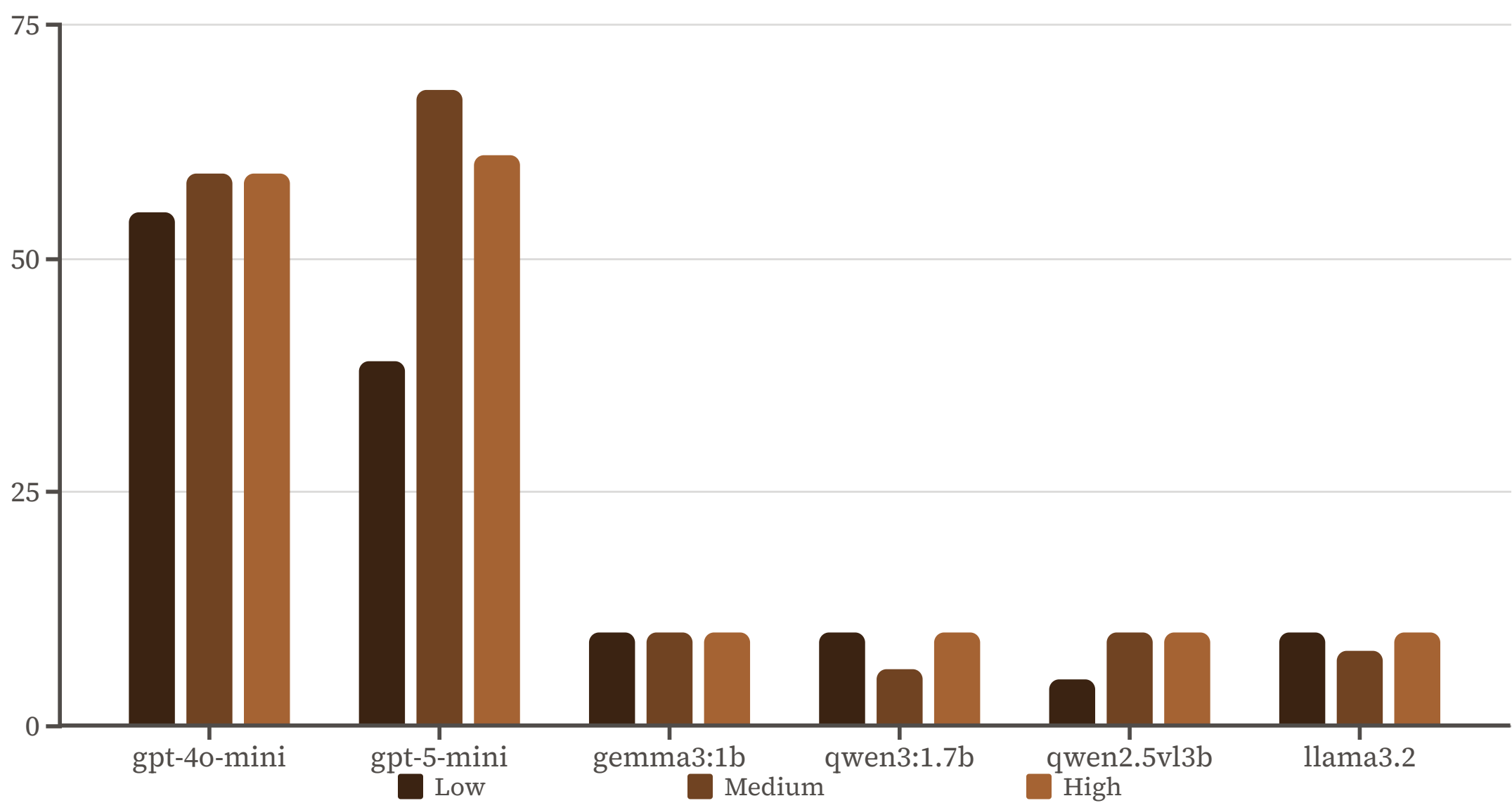


Natural Language to SQL: Enterprise Large Language Model Benchmarking Analysis

Comprehensive evaluation of Large Language Models for executive data query generation, assessing performance across complexity tiers from routine requests to advanced analytical queries.



LLM Performance Comparison by Query Complexity



Executive Insights

- **GPT-5-mini** leads in medium and high complexity queries, demonstrating 8% and 3% improvement over runner-up respectively.
- **GPT-4o-mini** shows consistent mid-tier performance across all complexity levels with 59 points at medium and high tiers.
- Open-source models including **Gemma**, **Qwen**, and **Llama** variants significantly underperform enterprise-grade solutions—minimum 49 points behind in medium complexity tier.

Query review scores reflect accuracy in SQL generation, handling of edge cases, and execution readiness. Higher scores indicate improved reliability for production deployment, particularly critical in high-complexity analytical workloads requiring advanced join logic and subquery handling.